

Relational Database

↳ Stores the in form of table.

↳ Mathematical term.

↓
Set

Table ⇒ 2D structure.

↳ Combination of Rows & Columns.

↓ ↓
Horizontal Vertical

Students

	↓	↓	↓	↓	↓
	id	name	phone	email	City
Row-1 →	1	Harold	1234	a@g.com	Delhi
Row-2 →	2	Kalpesh	7158	xyp@—	Ahmedabad
Row-3 →	2	Kalpesh	7158	xyp@—	Ahmedabad
→	—	—	—	—	—

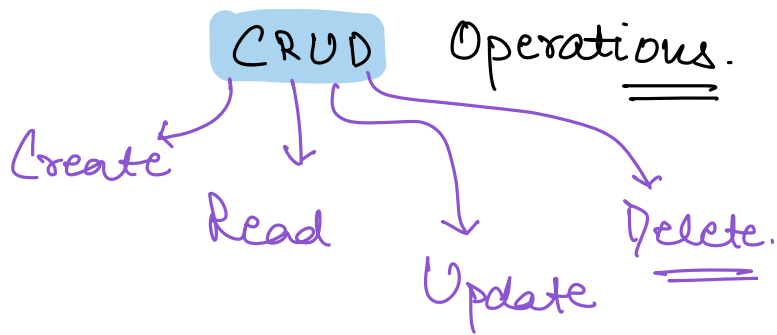
each row is 1 user.

Each column has a datatype



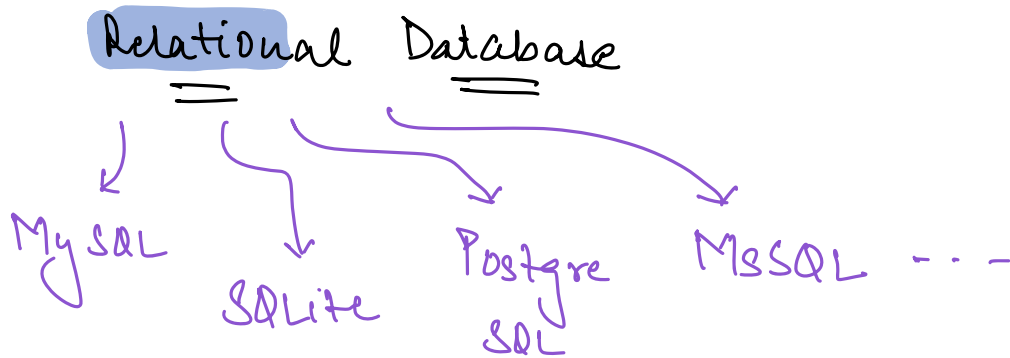
```
graph TD; A[datatype] --> B[int]; A --> C[String]; A --> D["..."];
```

DB enforces these datatype after every write.



SQL.

↳ Structured Query Language.



SQL : Case insensitive.

CRUD Operations using SQL.

Data Types in SQL

- INTEGER
- TEXT
- VARCHAR
- BOOLEAN
- DATE

1. CREATE Table.

table name
↓
Create table **students** (
 id INTEGER **UNIQUE** **NOT NULL** ,
 name VARCHAR(50) ,
 email VARCHAR(50) **UNIQUE** **NOT NULL** ,
 age INTEGER CHECK (age >= 0)
)

Primary Key (PK)

Any column of the table which is unique & not null.

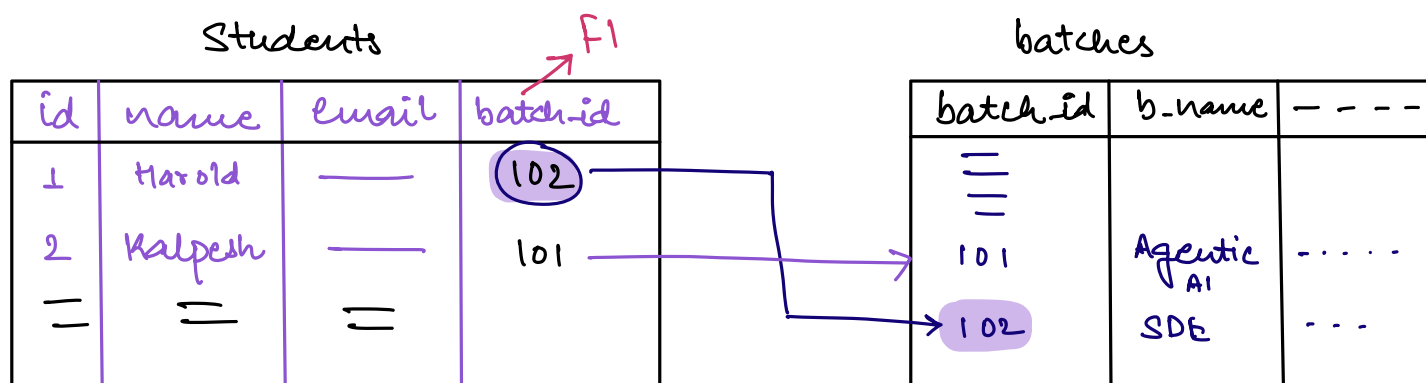
⇒ One table can have only 1 PK.

⇒ PK should never be changed.

Foreign Key.

→ A column that references the primary key of another table.

→ It creates a link b/w the two tables.



→ ON CASCADE DELETE



Deletes the dependent row when the reference row gets deleted.

SQL Queries

- INSERT
- SELECT → Retrieve data from table.
- UPDATE
- DELETE

```
use masai_june26;
```

```
create table IF NOT EXISTS students (  
    id integer unique not null,  
    name varchar(50),  
    email varchar(50) UNIQUE,  
    age integer check(age>=0)  
);
```

```
select * from students;
```

```
insert into students  
values (1, 'Harold', 'harold@gmail.com', 20);
```

```
insert into students  
values (2, 'Kalpesh', null, 21);
```

```
insert into students  
values (3, 'Jyotirmay', 'jyotirmay@gmail.com', 22);
```

```
-- Error Code: 1048. Column 'id' cannot be null  
-- Error Code: 3819. Check constraint 'students_chk_1' is violated.
```

```
insert into students  
values (3, 'Siddhartha', 'Siddhartha@gmail.com', 23);
```

```
-- Error Code: 1062. Duplicate entry '3' for key 'students.id'
```

```
create table users (  
    id integer primary key auto_increment,  
    name varchar(50),  
    email varchar(50) UNIQUE,  
    age integer check(age>=0)  
);
```

```
select * from users;
```

```
insert into users  
(name, email, age)  
values ('Ruchi', 'Ruchi@gmail.com', 22);
```

```
create table batches (  
    batch_id integer primary key,  
    batch_name varchar(100)  
);
```

```
insert into batches  
values (1, 'Agentic AI');
```

```
insert into batches
values (2, 'SDE');
```

```
insert into batches
values (3, 'Data Engineering');
```

```
insert into batches
values (4, 'AIML');
```

```
select * from batches;
```

```
drop table students;
```

```
create table IF NOT EXISTS students (
    id integer unique not null,
    name varchar(50),
    email varchar(50) UNIQUE,
    age integer check(age>=0),
    batch_id int,

    constraint fk_batch_id
    foreign key (batch_id)
    references batches(batch_id) ON DELETE CASCADE
);
```

```
insert into students
values (1, 'Harold', 'Harold@gmail.com', 20, 1);
```

```
insert into students
values (2, 'Ruchi', 'ruchi@gmail.com', 21, 2);
-- Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails
(masai_june26`.`students`, CONSTRAINT `fk_batch_id` FOREIGN KEY (`batch_id`) REFERENCES
`batches` (`batch_id`))
```

```
insert into students
values (3, 'Jyotirmay', 'Jyotirmay@gmail.com', 21, 3);
```

```
insert into students
(id, name, age, batch_id)
values (4, 'Shelin', 22, 3);
```

```
delete from students;
```

```
-- Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE
that uses a KEY column. To disable safe mode, toggle the option in Preferences -> SQL Editor and
reconnect.
```

```
update students set name = 'Harold Bright';
-- Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE
that uses a KEY column. To disable safe mode, toggle the option in Preferences -> SQL Editor and
```

reconnect.

-- Error Code: 1364. Field 'id' doesn't have a default value

select * from students;

select name, email from students;

select * from students where id = 1;

select * from students where email = 'abc@gmail.com';

delete from batches where batch_id = 2;

delete from batches;

-- Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE that uses a KEY column. To disable safe mode, toggle the option in Preferences -> SQL Editor and reconnect.

-- disable the safe update mode

SET SQL_SAFE_UPDATES = 0;

-- Error Code: 1451. Cannot delete or update a parent row: a foreign key constraint fails
(masai_june26`.`students`, CONSTRAINT `fk_batch_id` FOREIGN KEY (batch_id) REFERENCES
`batches` (batch_id))

-- Error Code: 1451. Cannot delete or update a parent row: a foreign key constraint fails
(masai_june26`.`students`, CONSTRAINT `fk_batch_id` FOREIGN KEY (batch_id) REFERENCES
`batches` (batch_id))

-- SELECT, INSERT, UPDATE, DELETE